



**ΠΑΝΕΠΙΣΤΗΜΙΟ ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
ΣΧΟΛΗ ΜΗΧΑΝΙΚΩΝ
ΤΜΗΜΑ ΜΗΧΑΝΙΚΩΝ ΠΛΗΡΟΦΟΡΙΚΗΣ ΚΑΙ
ΥΠΟΛΟΓΙΣΤΩΝ**

ΕΡΓΑΣΤΗΡΙΟ

ΒΑΣΕΙΣ ΔΕΔΟΜΕΝΩΝ II

ΕΡΓΑΣΙΑ 4

**ΦΙΛΙΠΠΟΣ ΦΙΛΙΠΠΑΤΟΣ
ΑΜ:19390242**

ΕΚΠ. ΕΤΟΣ 2024-2025

1. **Συνδεθείτε** στην MySQL του συστήματός σας με όποιον από τους προαναφερόμενους τρόπους επιθυμείτε.

```
filip@SOPPILIF c:\xampp
# mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 19
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

2. Ελέγξτε αν υπάρχει ΒΔ με την ονομασία **personnel**.

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| personnel |
| phpmyadmin |
| test |
+-----+
6 rows in set (0.001 sec)
```

3. Επιλέξτε την ΒΔ personnel για **χρήση**.

```
MariaDB [(none)]> use personnel;
Database changed
```

4. Δημιουργήστε τον trigger με όνομα dept_update που αφυπνίζεται από το γεγονός "BEFORE UPDATE ON".

```
MariaDB [personnel]> delimiter //
```

```
MariaDB [personnel]> CREATE TRIGGER dept_update
```

```
  -> BEFORE UPDATE ON dept
```

```
  -> FOR EACH ROW
```

```
  -> BEGIN
```

```
  -> SET NEW.dname = UPPER(NEW.dname);
```

```
  -> END;
```

```
  -> //
```

```
Query OK, 0 rows affected (0.014 sec)
```



```
MariaDB [personnel]> delimiter ;
```

```
MariaDB [personnel]> UPDATE dept SET dname = 'Operations' WHERE deptno=70;
```

```
Query OK, 1 row affected (0.011 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```



```
MariaDB [personnel]> select * from dept;
```

DEPTNO	DNAME	LOC
50	SALES	ATHENS
60	ACCOUNTING	ATHENS
70	OPERATIONS	VOLOS

```
3 rows in set (0.000 sec)
```

5. Στον πίνακα Department θα προσθέσουμε τη στήλη no_of_employees (αριθμός υπαλλήλων που έχει το τμήμα).

```
ALTER TABLE department ADD (no_of_employees INT);
```

```
MariaDB [personnel]> ALTER TABLE dept ADD (no_of_employees INT);
```

```
Query OK, 0 rows affected (0.028 sec)
```

```
Records: 0  Duplicates: 0  Warnings: 0
```



```
MariaDB [personnel]> select * from dept;
```

DEPTNO	DNAME	LOC	no_of_employees
50	SALES	ATHENS	NULL
60	ACCOUNTING	ATHENS	NULL
70	OPERATIONS	VOLOS	NULL

```
3 rows in set (0.082 sec)
```

6. Στον πίνακα Department θα ενημερώσουμε τη στήλη no_of_employees (αριθμός υπαλλήλων που έχει το τμήμα).

```
UPDATE department
SET no_of_employees =
    (SELECT COUNT(*)
     FROM employee
     WHERE employee.deptno = department.deptno);
```

```
MariaDB [personnel]> UPDATE dept
-> SET no_of_employees =
-> (SELECT COUNT(*)
-> FROM emp
-> WHERE emp.deptno = dept.deptno);
Query OK, 3 rows affected (0.005 sec)
Rows matched: 3  Changed: 3  Warnings: 0
```

```
MariaDB [personnel]> select * from dept;
```

DEPTNO	DNAME	LOC	no_of_employees
50	SALES	ATHENS	4
60	ACCOUNTING	ATHENS	4
70	OPERATIONS	VOLOS	0

```
3 rows in set (0.001 sec)
```

7. Στον πίνακα Trigger που αφυπνίζεται από το γεγονός AFTER INSERT ON

```
delimiter //

CREATE TRIGGER emp_insert

AFTER INSERT ON employee

FOR EACH ROW

BEGIN

UPDATE department

SET no_of_employees = IFNULL(no_of_employees, 0) + 1

WHERE deptno= NEW.deptno;

END;

//

delimiter ;
```

```
MariaDB [personnel]> UPDATE dept
-> SET no_of_employees =
-> (SELECT COUNT(*)
-> FROM emp
-> WHERE emp.deptno = dept.deptno);
Query OK, 3 rows affected (0.005 sec)
Rows matched: 3  Changed: 3  Warnings: 0
```

```
MariaDB [personnel]> select * from dept;
```

DEPTNO	DNAME	LOC	no_of_employees
50	SALES	ATHENS	4
60	ACCOUNTING	ATHENS	4
70	OPERATIONS	VOLOS	0

```
3 rows in set (0.001 sec)
```

8. Δοκιμή

```
INSERT INTO employee VALUES(7985, 'CLARKE', 10);

SELECT * FROM employee;

SELECT * FROM department;
```

```
MariaDB [personnel]> INSERT INTO emp VALUES (7985, 'CLARKE', 10, 50, NULL);
Query OK, 1 row affected (0.008 sec)
```

```
MariaDB [personnel]> select * from emp;
```

EMPNO	NAME	JOBNO	DEPTNO	COMM
10	CODD	100	50	NULL
20	NAVATHE	200	50	450.00
30	ELMASRI	300	60	NULL
40	DATE	100	50	NULL
90	CLARKE	100	50	NULL
100	adams	200	60	NULL
110	NAVATHE	100	60	NULL
120	ELMASRI	100	60	NULL
7985	CLARKE	10	50	NULL

```
9 rows in set (0.001 sec)
```

```
MariaDB [personnel]> select * from dept;
```

DEPTNO	DNAME	LOC	no_of_employees
50	SALES	ATHENS	5
60	ACCOUNTING	ATHENS	4
70	OPERATIONS	VOLOS	0

```
3 rows in set (0.001 sec)
```

9. Επεξήγηση

1. Ο trigger «αφυπνίζεται» λόγω της συνθήκης AFTER INSERT ON employee
2. OLD.empno<--NULL, OLD.ename<--NULL, OLD.deptno<--NULL
3. NEW.empno<-7985, NEW.ename<-'NAVATHE', NEW.deptno<-10

Άρα η δήλωση

```
UPDATE department
```

```
SET no_of_employees = NVL(no_of_employees, 0) + 1
```

```
WHERE deptno= :NEW.deptno;
```

Είναι ισοδύναμη με τη δήλωση

```
UPDATE department
```

```
SET no_of_employees = NVL(no_of_employees, 0) + 1
```

```
WHERE deptno= 10;
```

10. Δημιουργήστε, δοκιμάστε και εξηγήστε τον trigger που αφυπνίζεται από το γεγονός AFTER DELETE ON

```
MariaDB [personnel]> delimiter //
```

```
MariaDB [personnel]> CREATE TRIGGER emp_delete
```

```
  -> AFTER DELETE ON emp
```

```
  -> FOR EACH ROW
```

```
  -> BEGIN
```

```
  -> UPDATE dept
```

```
  -> SET no_of_employees = IFNULL(no_of_employees,0) - 1
```

```
  -> WHERE deptno=OLD.deptno;
```

```
  -> END;
```

```
  -> //
```

```
Query OK, 0 rows affected (0.014 sec)
```

```
MariaDB [personnel]> delimiter ;
```

```
MariaDB [personnel]> DELETE FROM emp WHERE empno = 7985;
```

```
Query OK, 1 row affected (0.010 sec)
```

```
MariaDB [personnel]> select * from emp;
```

EMPNO	NAME	JOBNO	DEPTNO	COMM
10	CODD	100	50	NULL
20	NAVATHE	200	50	450.00
30	ELMASRI	300	60	NULL
40	DATE	100	50	NULL
90	CLARKE	100	50	NULL
100	adams	200	60	NULL
110	NAVATHE	100	60	NULL
120	ELMASRI	100	60	NULL

```
8 rows in set (0.001 sec)
```

```
MariaDB [personnel]> select * from dept;
```

DEPTNO	DNAME	LOC	no_of_employees
50	SALES	ATHENS	4
60	ACCOUNTING	ATHENS	4
70	OPERATIONS	VOLOS	0

```
3 rows in set (0.001 sec)
```

1. Ο trigger «αφυπνίζεται» λόγω της συνθήκης AFTER DELETE ON employee
2. Για τη διαγραφή DELETE FROM emp WHERE empno = 7985:
 - OLD.empno <- 7985
 - OLD.ename <- 'CLARKE'
 - OLD.deptno <- 10
3. NEW.empno <- NULL, NEW.ename <- NULL, NEW.deptno <- NULL
Άρα η δήλωση:

UPDATE department

SET no_of_employees = IFNULL(no_of_employees, 0) - 1

WHERE deptno = OLD.deptno;

Είναι ισοδύναμη με τη δήλωση:

UPDATE department

SET no_of_employees = IFNULL(no_of_employees, 0) - 1

WHERE deptno = 10;

11. Δημιουργήστε, δοκιμάστε και εξηγήστε τον trigger που αφυπνίζεται από το γεγονός AFTER UPDATE ON

```
MariaDB [personnel]> delimiter //
```

```
MariaDB [personnel]> CREATE TRIGGER emp_update
```

```
  -> AFTER UPDATE ON emp
```

```
  -> FOR EACH ROW
```

```
  -> BEGIN
```

```
  -> IF OLD.deptno != NEW.deptno THEN
```

```
  -> UPDATE dept
```

```
  -> SET no_of_employees = IFNULL(no_of_employees,0) -1
```

```
  -> WHERE deptno=OLD.deptno;
```

```
  -> UPDATE dept
```

```
  -> SET no_of_employees=IFNULL(no_of_employees, 0) +1
```

```
  -> WHERE deptno=NEW.deptno;
```

```
  -> END IF;
```

```
  -> END
```

```
  -> //
```

```
Query OK, 0 rows affected (0.015 sec)
```

```
MariaDB [personnel]> delimiter :
```

```
MariaDB [personnel]> SELECT * FROM DEPT;
```

DEPTNO	DNAME	LOC	no_of_employees
50	SALES	ATHENS	4
60	ACCOUNTING	ATHENS	4
70	OPERATIONS	VOLOS	0

```
3 rows in set (0.001 sec)
```

```
MariaDB [personnel]> UPDATE emp SET deptno=60 WHERE empno= 90;
```

```
Query OK, 1 row affected (0.014 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [personnel]> select * from emp;
```

EMPNO	NAME	JOBNO	DEPTNO	COMM
10	CODD	100	50	NULL
20	NAVATHE	200	50	450.00
30	ELMASRI	300	60	NULL
40	DATE	100	50	NULL
90	CLARKE	100	60	NULL
100	adams	200	60	NULL
110	NAVATHE	100	60	NULL
120	ELMASRI	100	60	NULL

```
8 rows in set (0.001 sec)
```

```
MariaDB [personnel]> SELECT * FROM DEPT;
```

DEPTNO	DNAME	LOC	no_of_employees
50	SALES	ATHENS	3
60	ACCOUNTING	ATHENS	5
70	OPERATIONS	VOLOS	0

```
3 rows in set (0.001 sec)
```

1. Ο trigger «αφυπνίζεται» λόγω της συνθήκης AFTER UPDATE ON employee
2. OLD.empno <- 90
OLD.ename <- CLARKE
OLD.deptno <- 50
3. NEW.empno <- 90
NEW.ename <- CLARKE
NEW.deptno <- 60

Άρα οι δηλώσεις:

```
IF OLD.deptno != NEW.deptno THEN
```

```
UPDATE department
```

```
SET no_of_employees = IFNULL(no_of_employees, 0) - 1
```

```
WHERE deptno = OLD.deptno;
```

```
UPDATE department
```

```
SET no_of_employees = IFNULL(no_of_employees, 0) + 1
```

```
WHERE deptno = NEW.deptno;
```

```
END IF;
```

Είναι ισοδύναμες με τις δηλώσεις:

```
IF [παλιό τμήμα] != 60 THEN
```

```
UPDATE department
```

```
SET no_of_employees = IFNULL(no_of_employees, 0) - 1
```

```
WHERE deptno =50;
```

```
UPDATE department
```

```
SET no_of_employees = IFNULL(no_of_employees, 0) + 1
```

```
WHERE deptno = 60;
```

```
END IF;
```

12. Παρουσίαση πληροφοριών για τους Triggers

```
/* see triggers */
DESCRIBE Information_schema.TRIGGERS;

SELECT TRIGGER_NAME, EVENT_MANIPULATION, TRIGGER_SCHEMA
FROM INFORMATION_SCHEMA.TRIGGERS
WHERE TRIGGER_SCHEMA = 'my_first_triggers_db'
ORDER BY TRIGGER_NAME;
```

```
MariaDB [personnel]> DESCRIBE Information_schema.TRIGGERS;
```

Field	Type	Null	Key	Default	Extra
TRIGGER_CATALOG	varchar(512)	NO		NULL	
TRIGGER_SCHEMA	varchar(64)	NO		NULL	
TRIGGER_NAME	varchar(64)	NO		NULL	
EVENT_MANIPULATION	varchar(6)	NO		NULL	
EVENT_OBJECT_CATALOG	varchar(512)	NO		NULL	
EVENT_OBJECT_SCHEMA	varchar(64)	NO		NULL	
EVENT_OBJECT_TABLE	varchar(64)	NO		NULL	
ACTION_ORDER	bigint(4)	NO		NULL	
ACTION_CONDITION	longtext	YES		NULL	
ACTION_STATEMENT	longtext	NO		NULL	
ACTION_ORIENTATION	varchar(9)	NO		NULL	
ACTION_TIMING	varchar(6)	NO		NULL	
ACTION_REFERENCE_OLD_TABLE	varchar(64)	YES		NULL	
ACTION_REFERENCE_NEW_TABLE	varchar(64)	YES		NULL	
ACTION_REFERENCE_OLD_ROW	varchar(3)	NO		NULL	
ACTION_REFERENCE_NEW_ROW	varchar(3)	NO		NULL	
CREATED	datetime(2)	YES		NULL	
SQL_MODE	varchar(8192)	NO		NULL	
DEFINER	varchar(189)	NO		NULL	
CHARACTER_SET_CLIENT	varchar(32)	NO		NULL	
COLLATION_CONNECTION	varchar(32)	NO		NULL	
DATABASE_COLLATION	varchar(32)	NO		NULL	

```
22 rows in set (0.044 sec)
```

```
MariaDB [personnel]> SELECT TRIGGER_NAME, EVENT_MANIPULATION, TRIGGER_SCHEMA
-> FROM INFORMATION_SCHEMA.TRIGGERS
-> WHERE TRIGGER_SCHEMA = 'my_first_triggers_db'
-> ORDER BY TRIGGER_NAME;
```

```
Empty set (0.037 sec)
```

13. Διαγραφή των Triggers

```
DROP TRIGGER dept_insert;
```

```
DROP TRIGGER dept_update;
```

```
.....
```

```
MariaDB [personnel]> DROP TRIGGER dept_insert;  
ERROR 1360 (HY000): Trigger does not exist  
MariaDB [personnel]> DROP TRIGGER dept_update;  
Query OK, 0 rows affected (0.017 sec)
```

```
MariaDB [personnel]> DROP TRIGGER emp_insert;  
Query OK, 0 rows affected (0.010 sec)
```

```
MariaDB [personnel]> DROP TRIGGER emp_update;  
Query OK, 0 rows affected (0.013 sec)
```

```
MariaDB [personnel]> DROP TRIGGER emp_delete;  
Query OK, 0 rows affected (0.004 sec)
```